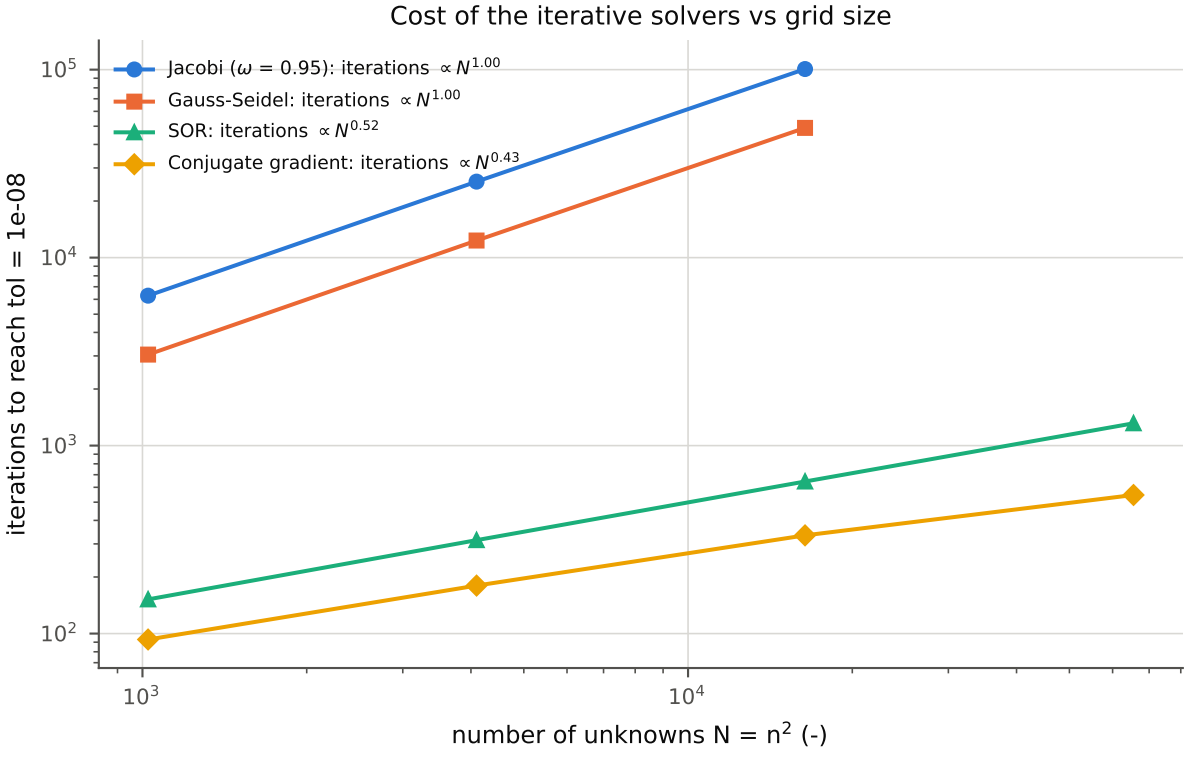
Cost of the iterative solvers vs grid size
Jacobi (ω = 0.95): iterations ∝ N^1.00
Gauss-Seidel: iterations ∝ N^1.00
SOR: iterations ∝ N^0.52
Conjugate gradient: iterations ∝ N^0.43
iterations to reach tol = 1e-08
number of unknowns N = n² (-)
10^5
10^4
10^3
10^2
10^3
10^4